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classmate

Date

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Ex 1: $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2), \mu \in \mathbb{R}, \sigma^2 \in \mathbb{R}_+$
 $\theta = (\mu, \sigma^2)$

NF-factorization:

$$g(\theta, T) = \exp \left[-\frac{1}{2\sigma^2} \left\{ \sum_{i=1}^n X_i^2 - 2\mu \sum_{i=1}^n X_i \right\} \right] \times p(\theta)$$

$$h(X) = 1$$

$\therefore T(X) = \left(\sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right)$ will be suff.

Q) Will it be complete?

A) Yes, using following result.

* Multiparametric Exponential Family of Distributions

$$X \sim p(\theta) \exp \{ C_1(\theta) T_1(X) + \dots + C_k(\theta) T_k(X) \} \cdot h(X)$$

$$\theta = (\theta_1, \dots, \theta_k), \theta \in H_0$$

- When we have $X_1, \dots, X_n$ i.i.d. having p.d.f./p.m.f. in an exponential family in the above form, a sufficient statistic for θ is:

$$S = S(X_1, \dots, X_n) = \sum_{i=1}^n (T_1(X_i), \dots, T_k(X_i))$$

- The natural parameter space is:

$$(\eta_1, \eta_2, \dots, \eta_k) = \{ (C_1(\theta), \dots, C_k(\theta)) : \theta \in H_0 \}$$

- For completeness, we need the natural parameter space to contain a rectangle w/ a non-empty interior

Ex2: Multinomial Distribution:

$$X = (Y_1, \dots, Y_k), \quad \theta = (p_1, \dots, p_{k-1}) \quad (p_1, \dots, p_{k-1})$$

$$\text{p.m.f.} \quad f(y_1, \dots, y_k) = \frac{n!}{\prod_{i=1}^k y_i!} \left(\prod_{i=1}^{k-1} p_i^{y_i} \right) (1 - p_1 - \dots - p_{k-1})^{y_k}$$

$$\text{conditions} \quad 0 \leq y_i \leq n, \quad \sum_{i=1}^k y_i = n$$

$$0 < p_i < 1, \quad \sum_{i=1}^k p_i = 1$$

Q) Is this in the multiparametric exponential family?

$$\begin{aligned} \text{A) } f(y_1, \dots, y_k) &= \frac{n!}{\prod_{i=1}^k y_i!} \exp \left[y_1 \log p_1 + \dots + y_{k-1} \log p_{k-1} \right. \\ &\quad \left. + (n - \sum_{i=1}^{k-1} y_i) \log (1 - p_1 - \dots - p_{k-1}) \right] \\ &= \frac{n!}{\prod_{i=1}^k y_i!} \exp \left[y_1 \log \left(\frac{p_1}{1 - p_1 - \dots - p_{k-1}} \right) + y_2 \log \left(\frac{p_2}{1 - p_1 - \dots - p_{k-1}} \right) + \dots \right. \\ &\quad \left. + y_{k-1} \log \left(\frac{p_{k-1}}{1 - p_1 - \dots - p_{k-1}} \right) + n \log (1 - p_1 - \dots - p_{k-1}) \right] \end{aligned}$$

~~≠~~

$$\therefore h(X) = \frac{n!}{\prod_{i=1}^k y_i!}, \quad p(\theta) = (1 - p_1 - \dots - p_{k-1})^n$$

$$C_i(\theta) = \frac{p_i}{1 - p_1 - \dots - p_{k-1}} \log \left(\frac{p_i}{1 - p_1 - \dots - p_{k-1}} \right), \quad T_i(X) = Y_i \text{ for } 1 \leq i \leq k-1$$

$\therefore X$ is in the family, and

$S = (T_1, \dots, T_{k-1})$ is a complete suff. stat.

Ex 3: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Unif}[\phi, \psi]$, $\theta = (\phi, \psi) \in \mathbb{R}^2$, $\phi < \psi$.

Joint distribution:

$$f(x_1, \dots, x_n | \phi, \psi) = \prod_{i=1}^n \frac{1}{\psi - \phi} \cdot \mathbb{I}[\phi < x_i < \psi]$$

$$= (\psi - \phi)^{-n} \cdot \left(\prod_{i=1}^n \mathbb{I}[\phi < x_i] \cdot \mathbb{I}[x_i < \psi] \right)$$

$$= (\psi - \phi)^{-n} \cdot \mathbb{I}[\phi < x_{(1)}] \cdot \mathbb{I}[x_{(n)} < \psi]$$

$\therefore S = (X_{(1)}, X_{(n)})$ is sufficient.

Q) Is S complete?

A) Joint distribution of $X_{(1)}, X_{(n)}$:

$$f(X_{(1)}, X_{(n)} | \phi, \psi) = \frac{n(n-1)}{(\psi - \phi)^n} (X_{(n)} - X_{(1)})^{n-2}, \quad \phi < X_{(1)} < X_{(n)} < \psi$$

Let g be a function, s.t.:

$$E_{\theta}(g(S)) = 0 \quad \forall \theta$$

$$\Rightarrow \int_{\phi}^{\psi} \int_{\phi}^{x_{(n)}} g(x_{(1)}, x_{(n)}) \frac{n(n-1)}{(\psi - \phi)^n} (x_{(n)} - x_{(1)})^{n-2} dx_{(1)} dx_{(n)} = 0 \quad \forall \theta$$

$$\Rightarrow \int_{\phi}^{\psi} \int_{\phi}^{x_{(n)}} g(x_{(1)}, x_{(n)}) \cdot (x_{(n)} - x_{(1)})^{n-2} dx_{(1)} dx_{(n)} = 0 \quad \forall \theta$$

Apparent. Apparently, in this case $g(X_{(1)}, X_{(n)}) \equiv 0 \quad \forall \theta$
 $\therefore S$ is also complete.

Ex 4: $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Unif}[\theta, \theta+1]$, $\theta \in \mathbb{R}$

$$\text{Joint dist}^n: f(x_1, \dots, x_n | \theta) = \prod_{i=1}^n \mathbb{I}[\theta < x_i < \theta+1]$$

$$= \prod_{i=1}^n \mathbb{I}[\theta < x_{(1)} < x_{(n)} < \theta+1]$$

By NF theorem, ~~$(X_{(1)}, X_{(n)})$~~ is ~~suff.~~ suff. for θ .
But it is not complete, because $X_{(n)} - X_{(1)}$ (a function of the sufficient statistic) is free of θ (#check)

$\therefore (X_{(1)}, X_{(n)})$ is an ancillary stat.

We see $\bar{X} - \frac{1}{2}$ is unb. est. for θ .

By RB, $\frac{X_{(1)} + X_{(n)}}{2}$ is a better unb. est. (#How?)

Q) Is it the UMVUE?

A) Apparently not (Idk Why tho, I asked Gemini)

Ex5: $X \sim N(\theta, 1)$, $(H) = \{\theta_0, \theta_1\}$

$H_0: \theta = \theta_0$ v/s $H_1: \theta = \theta_1$

What is a test?

$$X \rightarrow 0 \leq \phi(X) \leq 1$$

$\hookrightarrow$ Prob. of accepting $H_A \Leftrightarrow$ Rejecting H_0
Test funcⁿ.

Power of the test = $E_{\theta_1}(\phi(X)) = \text{Prob}_{\theta_1}\{\text{Rejecting } H_0\}$

Note: Type II error = 1 - Power of test

Size of the test = $E_{\theta_0}(\phi(X)) = \text{Prob}_{\theta_0}\{\text{Rejecting } H_0\}$
= Type I error.

Let's say $X \sim f_{\theta}(x) \rightarrow$ p.d.f. or p.m.f., $\Theta = \{\theta_0, \theta_1\}$

$$T(X) = \frac{f_{\theta_1}(X)}{f_{\theta_0}(X)}$$

$$f_{\theta}(x) = g(\theta, T) \cdot h(x)$$

$$h(x) = f_{\theta_0}(x)$$

$$g(\theta, T) = \begin{cases} T & \theta = \theta_1 \\ 1 & \theta = \theta_0 \end{cases}$$

$\therefore T$ is suff.